# MILITARY SPECIFICATION

## CHARGE, PROPELLING, 155MM: XM197

## LOADING, ASSEMBLING AND PACKING

### 1. SCOPE

1.1 This specification covers requirements, quality assurance provisons, and the preparation for delivery criteria for one type of propellant charge: XM197 for use with M454 projectile in M109A1 Howitzer.

### 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-223-Powder Black.
MIL-P-46699-Propellant, M1 for use in charge Propelling M2 (White Bag) for 8 Inch Howitzer

STANDARDS

MILITARY

MIL-STD-105-Sampling procedures and tables for inspection by attributes
MIL-STD-286-Propellants Solid Sampling, examination and testing
MIL-STD-1167-Ammunition Data Card
MIL-STD-1168-Lot Numbering of Ammunition
MIL-STD-1231-Surveillance Inspection of Industrial Stocks of Propellant in storage

DRAWINGS



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ORDNANCE CORPS

7548965-Automatic Air Test for loaded containers details
8849912-Cap, igniter protector(paper)Assembly
9272776-Propelling Charge Assembly
9272777-Body
9272778-Pad, Igniter
9272779-End
9272780-Strap, Tie
9274207-Flash Reducer Loading Assembly
9274208-Flash Reducer Assembly
9274209-Body, Flash Reducer
9275216-Packing and Marking for Container, Metal Ammo for Charge Propellant 155 mm: XM197 Single-Charge Pack
ET-9272776-Equipment Tabulation

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The charge shall comply with all requirements specified on the drawing(dwg) 9272776, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 Surveillance inspection of bulk propellant. A surveillance inspection of bulk propellant used in the charge assembly shall be performed in accordance with 4.4.1. If the bulk propellant has less than 1.2 percent by weight of stabilizer, the bulk propellant lot shall be rejected.

3.4. Moisture content of black powder. The moisture content of the black powder of the igniter charge and fash reducer charge shall not exceed 0.7 percent when tested as specified in 4.4.2.

3.5 Propellant and Black Powder. Each charge of propellant shall be weighed and check weighed to a tolerance of plus or minus 0.20 ounces (see 4.4.3). Each charge of black powder shall be weighed and check weighed to a tolerance of plus or minus 0.10 ounces.

3.6 Air pressure test. Each container shall be subjected to an air pressure test. The container shall maintain a pressure of 3 plus or minus 0.1 pounds per square inch for a minimum of 15 seconds without indication of leakage. When the machine for testing metal containers (see drawing 7548965) is used, the requirement for maintaining pressure shall be reduced to a minimum of five seconds. After completion of the air pressure test, the plug for the air



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test hole shall be seated with a torque of 5 plus 07 foot pounds. The air pressure test shall be conducted in accordance with test procedure 4.4.4.

3.7 Workmanship.

3.7.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.7.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.7.3 Burrs. Parts shall be free from burrs.

3.7.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.7.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.7.6 Stitching. Stitching shall not be missing where required. Stitching shall be free from any breaks or skipped areas. Tight stitching which breaks when the material is pulled to a flat surface or specified contour, shall be rejected.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. Not applicable

4.3 Quality Conformance inspection

4.3.1 Lot formation. Lot formation shall be in accordance with MIL-STD-1168. Changes in lot numbers shall be recorded on Ammunition Data Cards in accordance with MIL-STD-1167 and MIL-STD-1168.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table I

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and Table II), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.



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Table I
Major Characteristics

| Lot Size | Sample Size | Lot Size | Sample Size | Lot Size | Sample Size | Lot Size | Sample Size |
|---|---|---|---|---|---|---|---|
| 1-72 | all | 140-141 | 124 | 256-259 | 177 | 541-549 | 230 |
| 73 | 72 | 142 | 125 | 260-262 | 178 | 550-559 | 231 |
| 74 | 73 | 143-144 | 126 | 263-265 | 179 | 560-569 | 232 |
| 75 | 74 | 145 | 127 | 266-268 | 180 | 570-579 | 233 |
| 76 | 75 | 146-147 | 128 | 269-271 | 181 | 580-590 | 234 |
| 77 | 76 | 148-149 | 129 | 272-275 | 182 | 591-601 | 235 |
| 78 | 77 | 150 | 130 | 276-278 | 183 | 602-613 | 236 |
| 79 | 78 | 151-152 | 131 | 279-282 | 184 | 614-625 | 237 |
| 80 | 79 | 153-154 | 132 | 283-285 | 185 | 626-637 | 238 |
| 81-82 | 80 | 155-156 | 133 | 286-289 | 186 | 638-649 | 239 |
| 83 | 81 | 157 | 134 | 290-293 | 187 | 650-662 | 240 |
| 84 | 82 | 158-159 | 135 | 294-296 | 188 | 663-676 | 241 |
| 85 | 83 | 160-161 | 136 | 297-300 | 189 | 677-690 | 242 |
| 86 | 84 | 162-163 | 137 | 301-304 | 190 | 691-704 | 243 |
| 87 | 85 | 164-165 | 138 | 305-308 | 191 | 705-719 | 244 |
| 88-89 | 86 | 166-167 | 139 | 309-312 | 192 | 720-735 | 245 |
| 90 | 87 | 168 | 140 | 313-316 | 193 | 736-751 | 246 |
| 91 | 88 | 169-170 | 141 | 317-320 | 194 | 752-768 | 247 |
| 92 | 89 | 171-172 | 142 | 321-324 | 195 | 769-785 | 248 |
| 93 | 90 | 173-174 | 143 | 325-329 | 196 | 786-804 | 249 |
| 94-95 | 91 | 175-176 | 144 | 330-333 | 197 | 805-823 | 250 |
| 96 | 92 | 177-178 | 145 | 334-338 | 198 | 824-842 | 251 |
| 97 | 93 | 179-180 | 146 | 339-342 | 199 | 843-863 | 252 |
| 98 | 94 | 181-182 | 147 | 343-347 | 200 | 864-884 | 253 |
| 99-100 | 95 | 183-184 | 148 | 348-352 | 201 | 885-907 | 254 |
| 101 | 96 | 185-187 | 149 | 353-356 | 202 | 908-930 | 255 |
| 102 | 97 | 188-189 | 150 | 357-361 | 203 | 931-955 | 256 |
| 103 | 98 | 190-191 | 151 | 362-366 | 204 | 956-981 | 257 |
| 104-105 | 99 | 192-193 | 152 | 367-372 | 205 | 982-1008 | 258 |
| 106 | 100 | 194-195 | 153 | 373-377 | 206 | 1009-1036 | 259 |
| 107 | 101 | 196-197 | 154 | 378-382 | 207 | 1037-1066 | 260 |
| 108-109 | 102 | 198-200 | 155 | 383-388 | 208 | 1067-1098 | 261 |
| 110 | 103 | 201-202 | 156 | 389-393 | 209 | 1099-1131 | 262 |
| 111 | 104 | 203-204 | 157 | 394-399 | 210 | 1132-1166 | 263 |
| 112-113 | 105 | 205-207 | 158 | 400-405 | 211 | 1167-1204 | 264 |
| 114 | 106 | 208-209 | 159 | 406-411 | 212 | 1205-1243 | 265 |
| 115 | 107 | 210-212 | 160 | 412-417 | 213 | 1244-1285 | 266 |
| 116-117 | 108 | 213-214 | 161 | 418-424 | 214 | 1286-1330 | 267 |
| 118 | 109 | 215-217 | 162 | 425-430 | 215 | 1331-1378 | 268 |
| 119 | 110 | 218-219 | 163 | 431-437 | 216 | 1379-1429 | 269 |
| 120-121 | 111 | 220-222 | 164 | 438-443 | 217 | 1430-1483 | 270 |
| 122 | 112 | 223-224 | 165 | 444-450 | 218 | 1484-1542 | 271 |
| 123-124 | 113 | 225-227 | 166 | 451-457 | 219 | 1543-1605 | 272 |
| 125 | 114 | 228-230 | 167 | 458-465 | 220 | 1606-1673 | 273 |
| 126-127 | 115 | 231-232 | 168 | 466-472 | 221 | 1674-1747 | 274 |
| 128 | 116 | 233-235 | 169 | 473-480 | 222 | 1748-1827 | 275 |
| 129-130 | 117 | 236-238 | 170 | 481-488 | 223 | 1828-1914 | 276 |
| 131 | 118 | 239-241 | 171 | 489-496 | 224 | 1915-2010 | 277 |
| 132-133 | 119 | 242-243 | 172 | 497-504 | 225 | 2011-2115 | 278 |
| 134 | 120 | 244-246 | 173 | 505-513 | 226 | 2116-2232 | 279 |
| 135-136 | 121 | 247-249 | 174 | 514-521 | 227 | 2233-2361 | 280 |
| 137 | 122 | 250-252 | 175 | 522-530 | 228 | over 2361 | 298 |
| 138-139 | 123 | 253-255 | 176 | 531-540 | 229 | | |



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Table II
Minor Characteristics

| Lot Size | Sample Size | Lot Size | Sample Size | Lot Size | Sample Size | Lot Size | Sample Size |
|---|---|---|---|---|---|---|---|
| 1-16 | all | 28-29 | 23 | 52-56 | 31 | 132-156 | 39 |
| 17 | 16 | 30-31 | 24 | 57-62 | 32 | 157-191 | 40 |
| 18 | 17 | 32-34 | 25 | 63-69 | 33 | 192-244 | 41 |
| 19-20 | 18 | 35-36 | 26 | 70-76 | 34 | 245-334 | 42 |
| 21 | 19 | 37-40 | 27 | 77-86 | 35 | 335-519 | 43 |
| 22-23 | 20 | 41-43 | 28 | 87-97 | 36 | 520-1120 | 44 |
| 24-25 | 21 | 44-47 | 29 | 98-112 | 37 | over 1120 | 45 |
| 26-27 | 22 | 48-51 | 30 | 113-131 | 38 | | |

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-

4.3.2.1.2 Flash reducer loading assembly (see dwg 9274207) covering a detail of dwg 9272776.

| Categories | Characteristics | Method of Inspection |
|---|---|---|
| Critical: | None defined. | |
| Major: | | |
| | 101. Weight of potassium sulphate | Measure* |
| | 102. Stitching (see 3.7.6) | Visual/Manual |
| | 103. Assembly not damaged to the extent that potassium sulfide can escape | Visual |

*Determine weight by suitable scale.

| | | |
|---|---|---|
| Minor: | | |
| | 201. No seam or opening incompletely stitched | Visual |
| | 202. No thread broken | Visual |
| | 203. Cloth proper | Visual |
| | 204. Marking | Visual |
| | 205. Workmanship(see 3.7) | Visual |

4.3.2.1.3 Propelling charge assembly (see dwg 9272776) before assembly of cap igniter protector assembly.

| Categories | Characteristics | Method of Inspection |
|---|---|---|
| Critical: | | |
| | 1. Weight of propellant (see 3.5) | Measure* |
| | 2. Charge ignitor assembly not missing or insecure | Manual/Visual |

*Determine weight by suitable scale.



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Major:

| | | |
|---|---|---|
| 101. | Weight of black powder and propellant (see 3.5) | Measure* |
| 102. | Materials (see 3.1) | Visual |
| 103. | Diameter, max | Gage |
| 104. | Length, max | Gage |
| 105. | Stitching (see 3.7.6) | Visual/Manual |
| 106. | Assembly not damaged or defective to the extent black powder or propellant is escaping or likely to escape during subsequent handling | Visual |
| 107. | All components present | Visual |
| 108. | Color of cloth proper | Visual |

*Determine weight with suitable scale.

Minor:

| | | |
|---|---|---|
| 201. | Thread not broken | Visual |
| 202. | Straps present and properly located not loose, broken or insecurely tied | Visual |
| 203. | Marking | Visual |
| 204. | Workmanship(see 3.7) | Visual |

4.3.2.1.4 Propellant charge assembly (see dwg 9272776).

| Categories | Characteristics | Method of Inspection |
|---|---|---|

Critical: None defined.

Major:

| | | |
|---|---|---|
| 101. | Cap igniter protector assembly properly assembled and fits snugly | Visual/Manual |

Minor: None defined.

4.3.2.1.5 Packing and marking for container, metal, ammo, for charge propellant 155mm: XM197 signle pack (see dwg 9275216 and section 5).

| Categories | Characteristics | Method of Inspection |
|---|---|---|

Critical: None defined.

Major:

| | | |
|---|---|---|
| 101. | Pad in bottom of container | Visual |
| 102. | Charge wrapped inserted in container with igniter end of charge down in container | Visual |
| 103. | Pads and spacer added to insure tight pack | Visual |



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104. Torque on spider assembly.....Gage
105. Air pressure test (see 3.6).....Gage
106. Torque on test plug after air pressure test (see 4.4.4).....Gage
107. Car seal present and properly secured..Visual
108. Test strip present.....Visual
109. Materials (see 3.1).....Visual

Minor:

201. Marking.....Visual
202. Protective finish.....Visual
203. Workmanship (see 3.7).....Visual/Manual

4.3.3 Testing. All tests shall be performed. Tests need not be performed in the order specified.

| Test | Test Classification |
|---|---|
| a. Surveillance | Critical |
| b. Moisture content of black powder | Major |
| c. Air pressure | Major |
| d. Moisture content of wrapper and pads | Major |

Of these tests, a, b, and d are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Surveillance test (see 3.3). A sample or samples selcted at random from the bulk lot of powder shall be tested in accordance with 4.4.1. If the samples fail to comply with 3.3., the bulk lot of powder shall be rejected and not used for the charge assembly.

4.3.3.2 Moisture content of black powder (see 3.4). One ounce sample minimum of black powder selected at random from each lot used in 8 hours production or fraction thereof shall be subjected to this test (see 4.4.2). If any sample fails to comply with 3.4, the production represented by the sample shall be rejected.

4.3.3.3 Air pressure test (see 3.6). Each container shall be tested in accordance with 4.4.4. If any item fails to comply with 3.6, the item shall be rejected and removed from the lot.

4.3.3.4 Moisture content of wrappers and pads (see dwg 8865306). A four ounce sample, minimum of each paper product utilized in each 8 hour production or fraction thereof shall be elected at random and subjected to this test (see 4.4.5). If any sample fails to comply with the requirements of dwg 8865306, the production represented by the sample shall be rejected.



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4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). ELs for assemblies are listed on the Equipment Tabulation, ET-9272776. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Surveillance test (see 3.3). Perform the surveillance test as specified in MIL-STD-1231. If the test conforms to the requirements of MIL-STD-1231 accept the bulk powder and use the charge assemblies. If the test fails to comply with the requirements select a sample at random from the bulk powder and test in accordance with MIL-STD-286, Method 201.4.

4.4.2 Moisture content of black powder (see 3.4). The contractor shall provide adequate production controls to assure compliance of each lot with requirement 3.4. The moisture content shall be determined in accordance with specification MIL-P-223, determination of moisture content, section 4.

4.4.3 Propellant and black powder (see-3.5). Each propellant charge or black powder charge shall be weighed and check-weighed. The check-weighing shall be accomplished independently of the original weighing using a different balance from that used to make the original weighing and, if performed manually, shall be performed by another operator.

4.4.4 Air pressure test (see 3.6). The test plug shall be removed from the container after propelling charge has been packed. The container shall be pressurized to 3.0 $\pm$ 0.3 pounds per square inch gage as indicated by a pressure measuring device which has a resolution of 0.1 pounds per square inch. The air supply shall be shut off. An indicated



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pressure drop in excess of 0.1 pounds per square inch. The air supply shall be shut off. An indicated pressure drop in excess of 0.1 pounds per square inch, on the pressure measuring device from the original reading when the air is shut off after a minimum 15 seconds time interval shall be cause for noncompliance with requirements in 3.6. When the machine for air testing metal containers is used (see dwg 7548965), the minimum time interval shall be set at 5 seconds. After completion with a torque of 5 to 7 foot pounds of the test, the test plug shall be assembled to the container after coating threads of plug and hole with graphite grease.

4.4.5 Paper products (see 4.3.3.4). The contractor shall provide adequate production controls to assure the compliance of each lot with the specified requirements. The moisture content of the paper products shall be determined by obtaining a sample of approximately 60 grams of the material from the sample provided and accurately determine the exact weight of the sample to .0001 gram and record. Place sample in a moisture teller (Harry W. Dietest Co. of Detroit, Michigan or an approved equal) and dry for 30 plus 1 minutes. After drying determine the weight of the dried sample to .0001 gram and record the weight determined. Again dry the sample in the moisture teller of 15 plus 1 minute and determine the weight of the dried sample to .0001 grams. If the difference in the first dried sample and the second dried weight is 0.2 percent or less, the sample shall be considered dried, otherwise continue 15 plus 1 minute drying and weighing until the 0.2 percent or less difference level is reached. The loss in weight shall be calculated as follows:

$$\text{Percent Moisture} = \frac{100\ (A-B)}{A}$$

Where A = weight of sample before drying
B = weight of sample after drying

The percentage moisture of the sample shall comply with the requirements on dwg 8865306.

## 5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The charge shall be packaged in accordance with dwg 9275216 using the specified moisture barrier.

5.1.2 Level C. The charge shall be packaged in accordance with dwg 9275216except that the moisture barrier may be omitted.



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5.2 Packing.

5.2.1 Level A and C. The charge shall be packed in accordance with dwg 9275216.

5.3 Marking. Marking shall be as specified in accordance with dwg. 9275216.

5.4 Data cards. Data cards shall be prepared in accoreance with MIL-STD-1167.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 3.4.3.1 and 5.4). Data specified in 3.4.3.1 and 5.4 will be delivered as identified on a numbered DD Form 1664 or specification when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-QA-I.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 1145 - A007



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-C-48009(MU) Charge, Propellant: XM197

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 65 1426 REPLACES EDITION OF 1 OCT 54 WHICH MAY BE USED

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